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Skill 0.1 Exercise 1

How are functions in Python similar to functions in mathematics? How are they different?

Both use inputs to produce results. Math functions map each input to one output; Python functions run reusable instructions and may return a value.

Skill 0.2 Exercise 1

Refer to the code below.

```
print("Python is awesome!")
```

What is the name of the function? What is the argument?

Function name: print

Argument: "Python is awesome!"

Refer to the code below.

```
print
```

Does this code invoke the print function? How do you know?

No. Without parentheses, print is referenced but not invoked.

Skill 0.3 Exercise 1

Python has many built-in functions. The signature for the round function is shown below. A function signature defines how to use a function. For example, according to the signature below, the round function can accept 1 or 2 arguments. The first argument is the number to be rounded. The second number is separated by a comma and specifies the number of digits after the decimal you want to round the number to. The second parameter is optional and if not provided the first argument will be rounded to the nearest whole number. Below are some examples,

round(*number*, [*ndigits*])

Function call	Value returned
round(5.89898, 1)	5.9
round(5.89898)	6
round(7)	7
round(10, 2)	10

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Based on the signature, indicate whether each invocation of <i>round</i> is legal or illegal, along with the value returned.		
Call	Legal/Illegal	Value returned
round(5.1)	Legal	5
round(5.109, 2)	Legal	5.11
round(5.1, 2, 3)	Illegal (3 arguments)	No value
round()	Illegal (missing argument)	No value
round(5.1, .1)	Illegal (.1 is not an integer)	No value

Skill 0.4 Exercise 1		
For each of the following print invocations, indicate whether the statement is legal or illegal. If it is legal, indicate the output.		
Code	Legal/Illegal	Output
print("I")	Illegal	No output (SyntaxError)
print("Love") print("To code")		
print("I") print print("Love") print("To Code")	Legal	I Love To Code
print("I") print() print("Love") print("To Code")	Legal	I Love To Code

Skill 0.5 Exercise 1
Indicate the output for the following statements
<pre>print("The itsy bitsy spider\nclimbed up the waterspout.") print() print("Down came the rain\nand washed the spider out.")</pre>
The itsy bitsy spider climbed up the waterspout. Down came the rain and washed the spider out.

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Skill 0.6 Exercise 1

Indicate the output for the following print statements.

```
print("My name is", "Python.")  
print("Monty Python.")
```

My name is Python.
Monty Python.

Skill 0.7 Exercise 1

Indicate the output for the following print statements,

```
print("My", "name", "is", sep="_", end="*")  
print("Monty", "Python.", sep="*", end="*\n")
```

My_name_is*Monty*Python.*

Set 0 Final Synthesis

Write one print() invocation that uses three positional arguments, sep, and end to produce the exact output shown.

Data|Science|Python!

Your code below

```
print("Data", "Science", "Python", sep="|", end="!")
```